

OpenStax Calculus Volume 2

Section 6.1 Exercises: Power Series and Functions

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In the following exercises, state whether each statement is true, or give an example to show that it is false.

1. If $\sum_{n=1}^{\infty} a_n x^n$ converges, then $a_n x^n \rightarrow 0$ as $n \rightarrow \infty$.
2. $\sum_{n=1}^{\infty} a_n x^n$ converges at $x = 0$ for any real numbers a_n .
3. Given any sequence a_n , there is always some $R > 0$, possibly very small, such that $\sum_{n=1}^{\infty} a_n x^n$ converges on $(-R, R)$.
4. If $\sum_{n=1}^{\infty} a_n x^n$ has radius of convergence $R > 0$ and if $|b_n| \leq |a_n|$ for all n , then the radius of convergence of $\sum_{n=1}^{\infty} b_n x^n$ is greater than or equal to R .
5. Suppose that $\sum_{n=0}^{\infty} a_n (x - 3)^n$ converges at $x = 6$. At which of the following points might the series diverge? Use the fact that if $\sum a_n (x - c)^n$ converges at x , then it converges at any point closer to c than x .
 - (a) $x = 1$
 - (b) $x = 2$
 - (c) $x = 3$
 - (d) $x = 0$
 - (e) $x = 5.99$
 - (f) $x = 0.000001$
6. Suppose that $\sum_{n=0}^{\infty} a_n (x + 1)^n$ converges at $x = -2$. At which of the following points must the series also converge? Use the fact that if $\sum a_n (x - c)^n$ converges at x , then it converges at any point closer to c than x .
 - (a) $x = 2$
 - (b) $x = -1$
 - (c) $x = -3$
 - (d) $x = 0$
 - (e) $x = 0.99$
 - (f) $x = 0.000001$

In the following exercises, suppose that $|\frac{a_{n+1}}{a_n}| \rightarrow 1$ as $n \rightarrow \infty$. Find the radius of convergence for each series.

7. $\sum_{n=0}^{\infty} a_n 2^n x^n$
8. $\sum_{n=0}^{\infty} \frac{a_n x^n}{2^n}$
9. $\sum_{n=0}^{\infty} \frac{a_n \pi^n x^n}{e^n}$
10. $\sum_{n=0}^{\infty} \frac{a_n (-1)^n x^n}{10^n}$
11. $\sum_{n=0}^{\infty} a_n (-1)^n x^{2n}$
12. $\sum_{n=0}^{\infty} a_n (-4)^n x^{2n}$

In the following exercises, find the radius of convergence R and interval of convergence for $\sum a_n x^n$ with the given coefficients a_n .

13. $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$
14. $\sum_{n=1}^{\infty} (-1)^n \frac{x^n}{\sqrt{n}}$
15. $\sum_{n=1}^{\infty} \frac{nx^n}{2^n}$
16. $\sum_{n=1}^{\infty} \frac{nx^n}{e^n}$
17. $\sum_{n=1}^{\infty} \frac{n^2 x^n}{2^n}$
18. $\sum_{k=1}^{\infty} \frac{k^e x^k}{e^k}$
19. $\sum_{k=1}^{\infty} \frac{\pi^k x^k}{k^\pi}$
20. $\sum_{n=1}^{\infty} \frac{x^n}{n!}$
21. $\sum_{n=1}^{\infty} \frac{10^n x^n}{n!}$
22. $\sum_{n=1}^{\infty} (-1)^n \frac{x^n}{\ln(2n)}$

In the following exercises, find the radius of convergence of each series.

23. $\sum_{k=1}^{\infty} \frac{(k!)^2 x^k}{(2k)!}$
24. $\sum_{n=1}^{\infty} \frac{(2n)! x^n}{n^{2n}}$
25. $\sum_{k=1}^{\infty} \frac{k!}{1 \cdot 3 \cdot 5 \cdots (2k-1)} x^k$
26. $\sum_{k=1}^{\infty} \frac{2 \cdot 4 \cdot 6 \cdots 2k}{(2k)!} x^k$
27. $\sum_{n=1}^{\infty} \frac{x^n}{\binom{2n}{n}}$ where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
28. $\sum_{n=1}^{\infty} \sin^2(n) x^n$

In the following exercises, use the ratio test to determine the radius of convergence of each series.

29. $\sum_{n=1}^{\infty} \frac{(n!)^3}{(3n)!} x^n$

$$30. \sum_{n=1}^{\infty} \frac{2^{3n}(n!)^3}{(3n)!} x^n$$

$$31. \sum_{n=1}^{\infty} \frac{n!}{n^n} x^n$$

$$32. \sum_{n=1}^{\infty} \frac{(2n)!}{n^{2n}} x^n$$

In the following exercises, given that $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ with convergence in $(-1, 1)$, find the power series for each function with the given center a , and identify its interval of convergence.

$$33. f(x) = \frac{1}{x}; a = 1. \text{ (Hint: } \frac{1}{x} = \frac{1}{1-(1-x)} \text{.)}$$

$$34. f(x) = \frac{1}{1-x^2}, a = 0$$

$$35. f(x) = \frac{x}{1-x^2}, a = 0$$

$$36. f(x) = \frac{1}{1+x^2}, a = 0$$

$$37. f(x) = \frac{x^2}{1+x^2}, a = 0$$

$$38. f(x) = \frac{1}{2-x}, a = 1$$

$$39. f(x) = \frac{1}{1-2x}; a = 0.$$

$$40. f(x) = \frac{1}{1-4x^2}, a = 0$$

$$41. f(x) = \frac{x^2}{1-4x^2}, a = 0$$

$$42. f(x) = \frac{x^2}{5-4x+x^2}, a = 2$$

Use the next exercise to find the radius of convergence of the given series in the subsequent exercises.

$$43. \text{ Explain why, if } |a_n|^{1/n} \rightarrow r > 0, \text{ then } |a_n x^n|^{1/n} \rightarrow |x|r < 1 \text{ whenever } |x| < \frac{1}{r} \text{ and, therefore, the radius of convergence of } \sum_{n=1}^{\infty} a_n x^n \text{ is } R = \frac{1}{r}.$$

$$44. \sum_{n=1}^{\infty} \frac{x^n}{n^n}$$

$$45. \sum_{k=1}^{\infty} \left(\frac{k-1}{2k+3} \right)^k x^k$$

$$46. \sum_{k=1}^{\infty} \left(\frac{2k^2-1}{k^2+3} \right)^k x^k$$

$$47. \sum_{n=1}^{\infty} a_n x^n, \text{ where } a_n = (n^{1/n} - 1)^n$$

$$48. \text{ Suppose that } p(x) = \sum_{n=0}^{\infty} a_n x^n \text{ such that } a_n = 0 \text{ if } n \text{ is even. Explain why } p(x) = -p(-x).$$

$$49. \text{ Suppose that } p(x) = \sum_{n=0}^{\infty} a_n x^n \text{ such that } a_n = 0 \text{ if } n \text{ is odd. Explain why } p(x) = p(-x).$$

$$50. \text{ Suppose that } p(x) = \sum_{n=0}^{\infty} a_n x^n \text{ converges on } (-1, 1]. \text{ Find the interval of convergence of } p(Ax).$$

$$51. \text{ Suppose that } p(x) = \sum_{n=0}^{\infty} a_n x^n \text{ converges on } (-1, 1]. \text{ Find the interval of convergence of } p(2x - 1).$$

In the following exercises, suppose that $p(x) = \sum_{n=0}^{\infty} a_n x^n$ satisfies $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$ where $a_n \geq 0$ for each n . State whether each series converges on the full interval $(-1, 1)$, or if there is not enough information to draw a conclusion. Use the comparison test when appropriate.

52. $\sum_{n=0}^{\infty} a_n x^{2n}$
53. $\sum_{n=0}^{\infty} a_{2n} x^{2n}$
54. $\sum_{n=0}^{\infty} a_{2n} x^n$ (Hint: $x = \pm\sqrt{x^2}$.)
55. $\sum_{n=0}^{\infty} a_{n^2} x^{n^2}$ (Hint: Let $b_k = a_k$ if $k = n^2$ for some n ; otherwise, $b_k = 0$.)
56. Suppose that $p(x)$ is a polynomial of degree N . Find the radius and interval of convergence of $\sum_{n=1}^{\infty} p(n)x^n$.
57. [T] Plot the graphs of $\frac{1}{1-x}$ and of the partial sums $S_N = \sum_{n=0}^N x^n$ for $N = 10, 20, 30$ on the interval $[-0.99, 0.99]$. Comment on the approximation of $\frac{1}{1-x}$ by S_N near $x = -1$ and near $x = 1$ as N increases.
58. [T] Plot the graphs of $-\ln(1-x)$ and of the partial sums $S_N = \sum_{n=1}^N \frac{x^n}{n}$ for $N = 10, 50, 100$ on the interval $[-0.99, 0.99]$. Comment on the behavior of the sums near $x = -1$ and near $x = 1$ as N increases.
59. [T] Plot the graphs of the partial sums $S_N = \sum_{n=1}^N \frac{x^n}{n^2}$ for $N = 10, 50, 100$ on the interval $[-0.99, 0.99]$. Comment on the behavior of the sums near $x = -1$ and near $x = 1$ as N increases.
60. [T] Plot the graphs of the partial sums $S_N = \sum_{n=1}^N \sin(n)x^n$ for $N = 10, 50, 100$ on the interval $[-0.99, 0.99]$. Comment on the behavior of the sums near $x = -1$ and near $x = 1$ as N increases.
61. [T] Plot the graphs of the partial sums $S_N = \sum_{n=0}^N (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ for $N = 3, 5, 10$ on the interval $[-2\pi, 2\pi]$. Comment on how these plots approximate $\sin x$ as N increases.
62. [T] Plot the graphs of the partial sums $S_N = \sum_{n=0}^N (-1)^n \frac{x^{2n}}{(2n)!}$ for $N = 3, 5, 10$ on the interval $[-2\pi, 2\pi]$. Comment on how these plots approximate $\cos x$ as N increases.